

Week 02: Comprehensive Combinatorics

1 Exercises

Exercise 1. How many binary strings of length 8 are there? Among these binary strings of length 8, how many contain exactly three 0-bits?

Exercise 2. How many strings of length 5 can be formed from the characters in $U = \{a, b, c, d\}$? How many strings contain exactly one character 'a'? How many strings contain at least one character 'a'?

Exercise 3. 4 guests are invited by a host to sit in a salon set with 6 seats. How many ways can these 4 guests choose their seats?

Exercise 4. A parking lot has 10 consecutive empty spaces. There are 6 different cars that need to be parked in these spaces. How many ways are there to arrange these 6 cars into the 10 spaces?

A class has 20 students, including 12 females and 8 males. A committee of 5 students needs to be selected.

a) How many ways are there to select this committee?

b) How many ways are there if the committee must have exactly 3 females and 2 males?

Exercise 5. From the set of digits $\{1, 2, 3, 4, 5, 6, 7\}$, a 4-digit password with distinct digits is to be created. How many passwords satisfy the following conditions:

a) The password is an even number.

b) The password must contain the digit 5.

Exercise 6. A bakery has 5 different types of donuts. A customer wants to buy 8 donuts. How many ways can these 8 donuts be chosen (the order of selection does not matter and multiple donuts of the same type can be chosen)?

Exercise 7. Find the number of distinct strings that can be formed by rearranging the letters in the word COMBINATORICS.

Exercise 8. There are 7 different letters to be placed into 4 different mailboxes. How many ways can these letters be distributed if:

a) There are no restrictions.

b) The first mailbox must contain at least 2 letters.

Exercise 9. Find the number of ways to distribute 15 identical marbles into 4 different boxes such that:

a) Each box has at least 1 marble.

b) The first box has at most 5 marbles.

Exercise 10. Prove that:

$$\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$$

2 Programming

Exercise 1. Write a function to verify the correctness of Exercise 10 in the Exercises section.

Exercise 2. Write a function `sol_equation(N, K)` to calculate the number of non-negative integer solutions to the equation $x_1 + x_2 + \cdots + x_K = N$.

Exercise 3. Write a function `verify_pascal_identity(n, k)` to verify the validity of Pascal's Identity:

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Exercise 4. Write a function `calculate_multiset_permutation(counts)` to calculate the number of permutations of a multiset containing identical elements. Then, verify the result of Exercise 7 in the Exercises section.

Exercise 5. Given a positive integer N and a positive integer K . Write a function `count_restricted_partitions(N, K)` to calculate the number of ways to partition the number N into a sum of positive integers that satisfy two properties:

- Each element in the sum must not be greater than K .
- Elements are allowed to repeat.

END
